

EduDesk Architecture & SRS Proposal

Overview

This proposal outlines a production-scale educational desktop workspace integrating authorized learning platforms, AI assistance, multi-window workspace management, strong backend architecture, and enterprise security.

Document Volumes

1. SRS
2. System Design
3. Database Design
4. Security Architecture
5. AI System
6. Desktop Workspace
7. Browser Integration
8. Knowledge Index
9. Backend Microservices
10. DevOps

Diagrams

Context Diagram, DFD Level 0/1/2, ER Diagram, UML Class, Sequence, Activity, Component, Deployment, State Machine, Network, Threat Model, Security Architecture, API Flow, Knowledge Graph, RAG Pipeline and many more.

Technology Stack

Flutter desktop, Go/Rust/Java/Python microservices, PostgreSQL, Redis, Neo4j, Vector Database, Docker, Kubernetes, gRPC, REST, WebSockets.

Security

Zero Trust, RBAC, OAuth2, JWT, MFA, TLS, encrypted storage, audit logs, code signing, plugin sandboxing, secure IPC, monitoring.

Deliverable

A future complete document would target approximately 180–250 pages with professional diagrams, colorful formatting, UI mockups, and detailed specifications.